

Transformations and the Library of Relations (Part II) Homework

Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation. Draw the asymptote(s) on the graph as dotted lines and write their equation(s). If the transformation is a dilation, then identify the type of dilation.

$$\textcircled{1} \frac{1}{x-2} = y+4$$

$$\textcircled{2} y-3 = \frac{1}{x+1}$$

$$\textcircled{3} \frac{1}{x+3} = y+3$$

$$\textcircled{4} y+2 = 4^{x-1}$$

$$\textcircled{5} 3^{x-2} = y-5$$

$$\textcircled{6} y-3 = 5^{x+4}$$

$$\textcircled{7} \log_2(x-1) = y+3$$

$$\textcircled{8} y-4 = \log_3(x-4)$$

$$\textcircled{9} \log_4(x+2) = y+1$$

$$\textcircled{10} y = 4^{-x}$$

$$\textcircled{11} \log_2 X = -y$$

$$\textcircled{12} y = \frac{1}{-x}$$

$$\textcircled{13} y = 2^{5x}$$

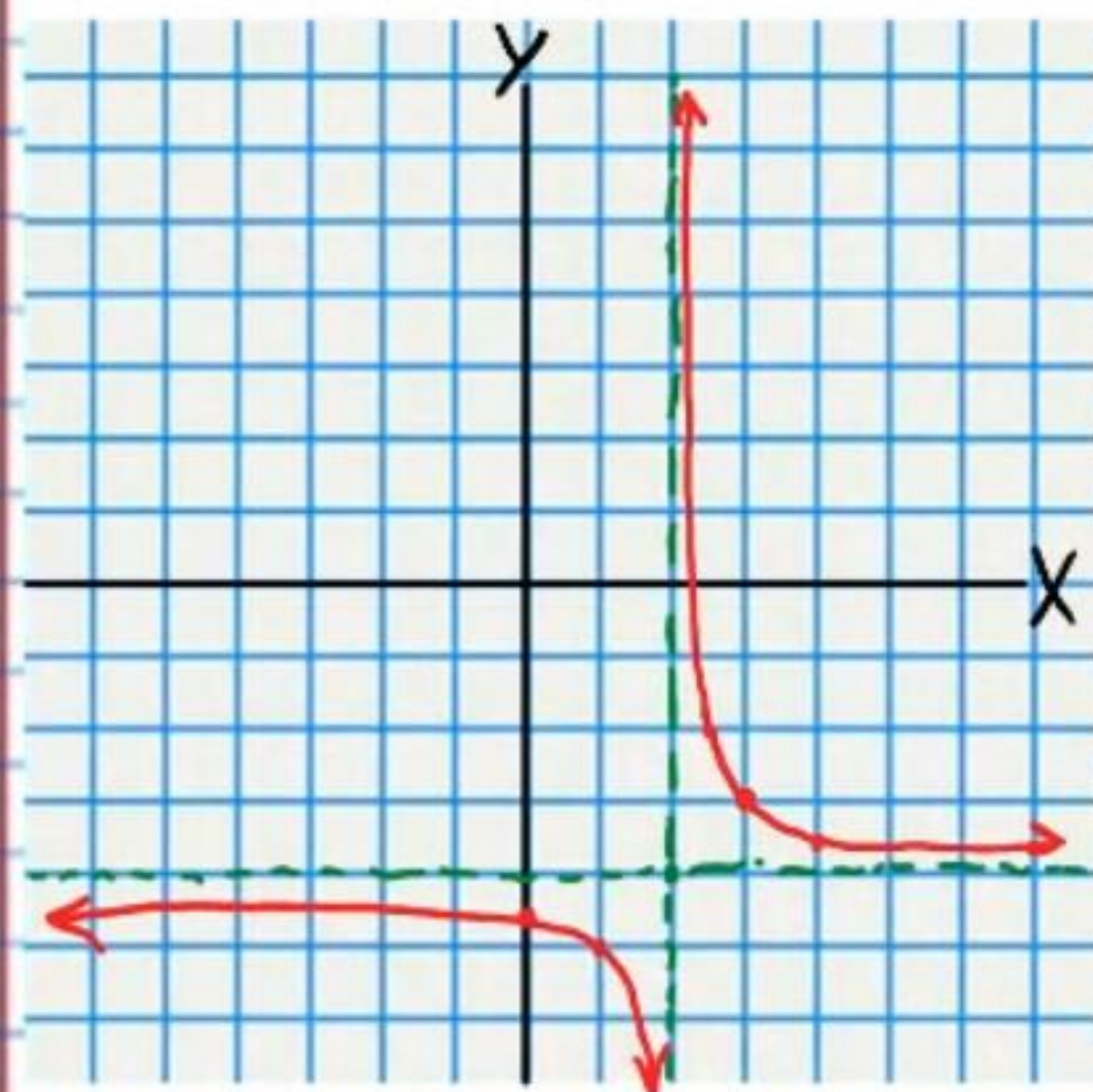
$$\textcircled{14} \frac{y}{3} = \frac{1}{x}$$

$$\textcircled{15} 4y = \log_2 X$$

Answers

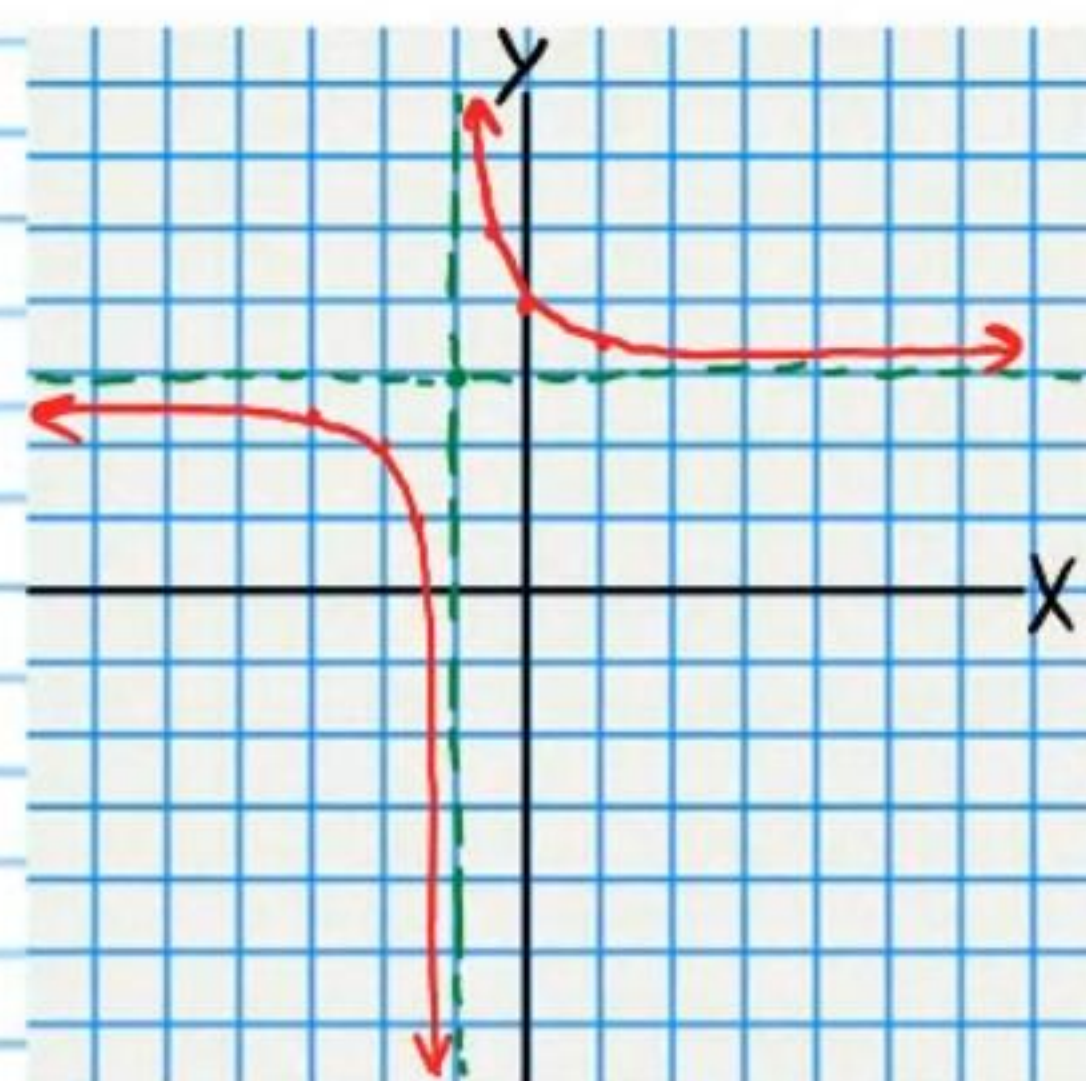
① Reciprocal, $\frac{1}{x} = y$

$x = 2, y = -4$



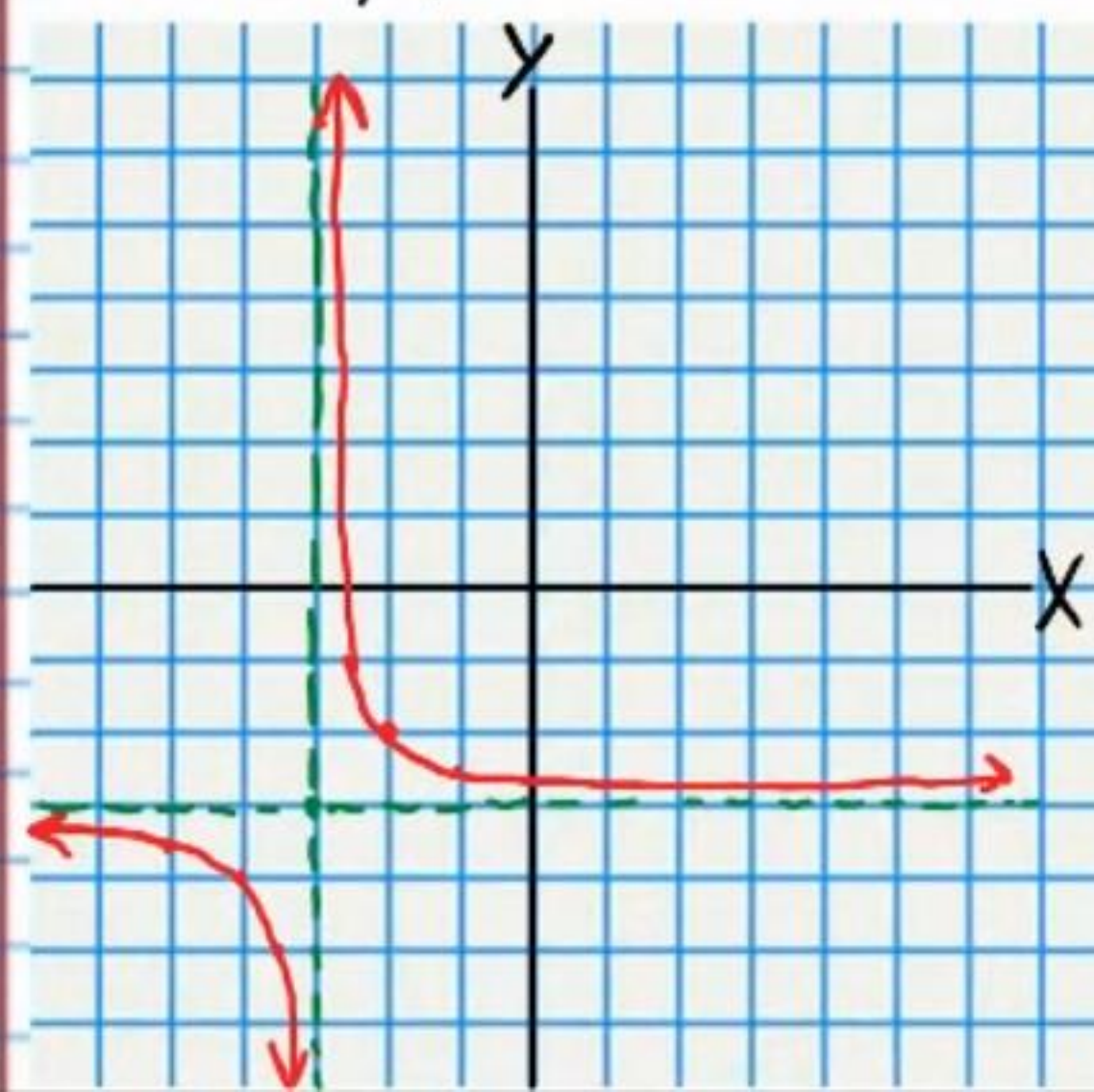
② Reciprocal, $y = \frac{1}{x}$

$x = -1, y = 3$



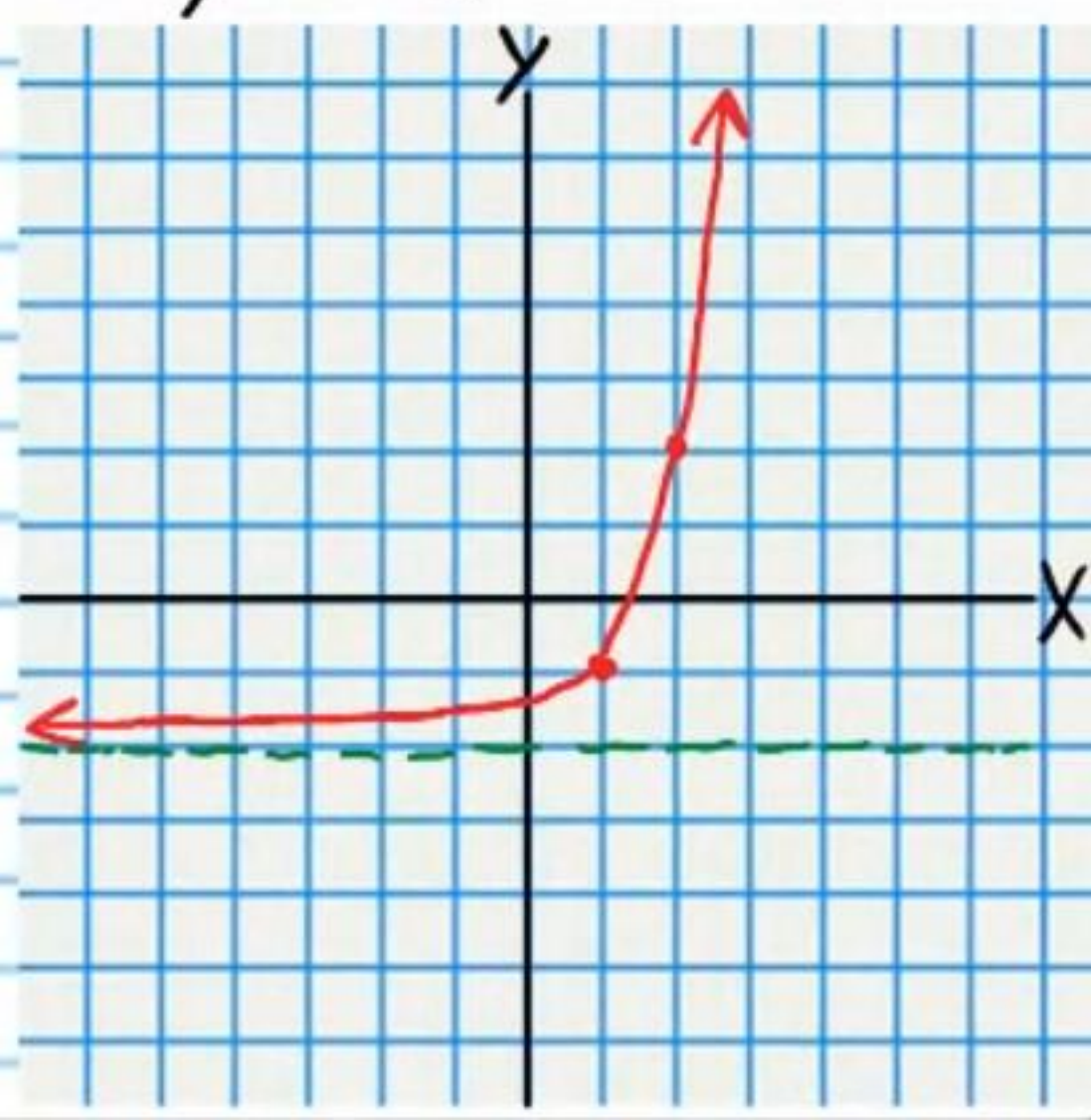
③ Reciprocal, $\frac{1}{x} = y$

$x = -3, y = -3$



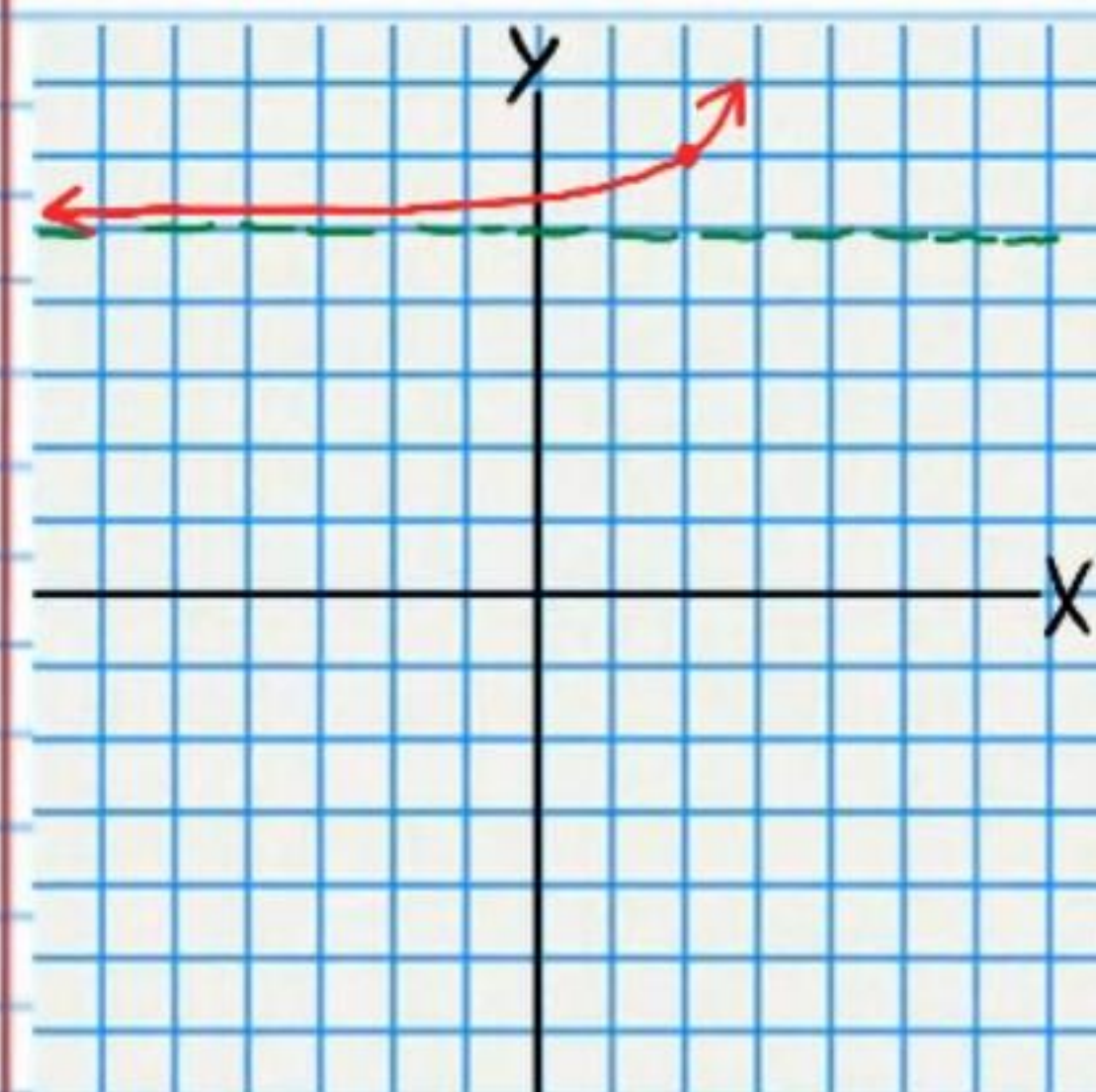
④ Exponential, $y = 4^x$

$y = -2$



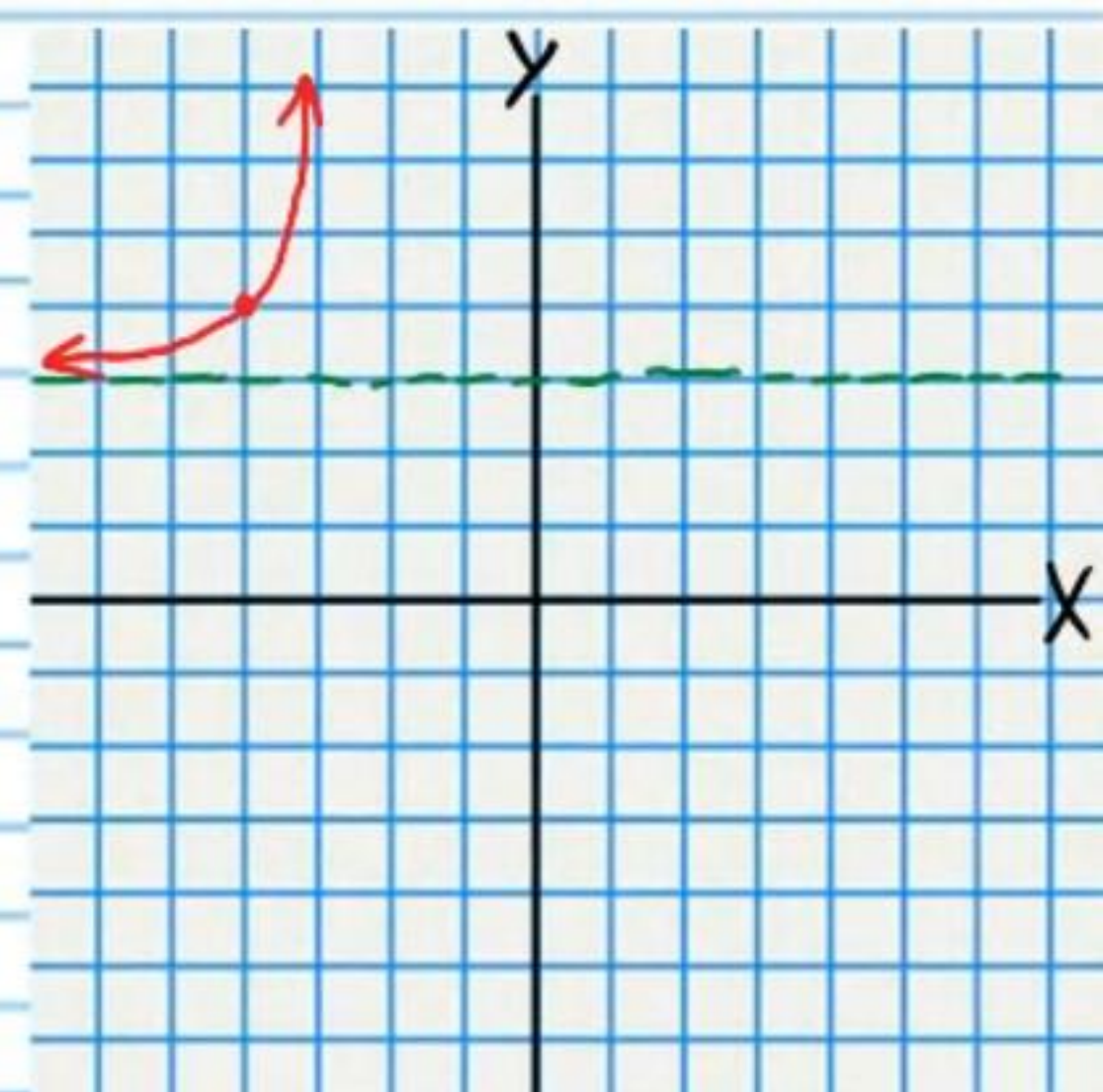
⑤ Exponential, $3^x = y$

$y = 5$



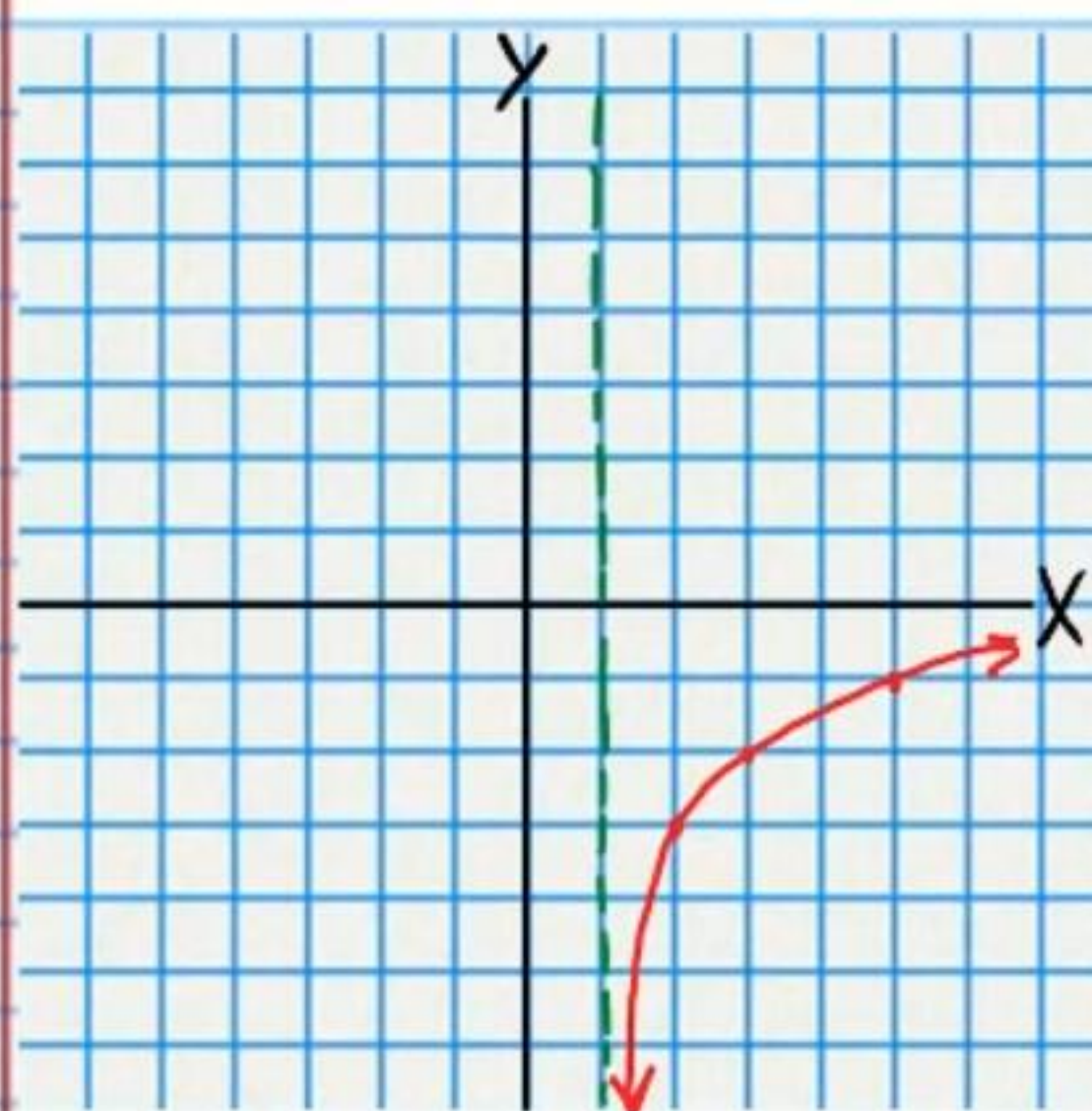
⑥ Exponential, $y = 5^x$

$y = 3$



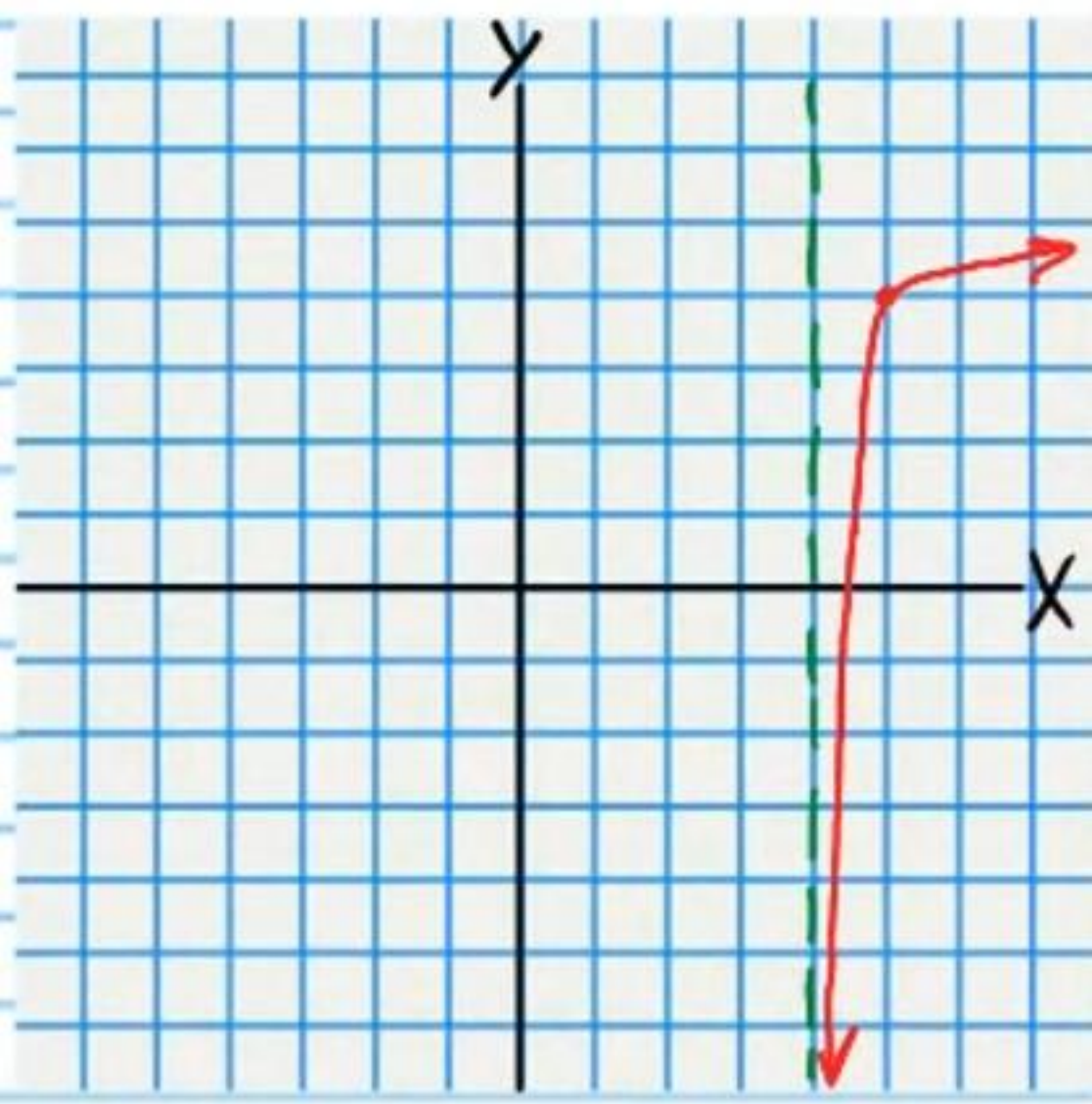
⑦ Logarithmic, $\log_2 x = y$

$x = 1$

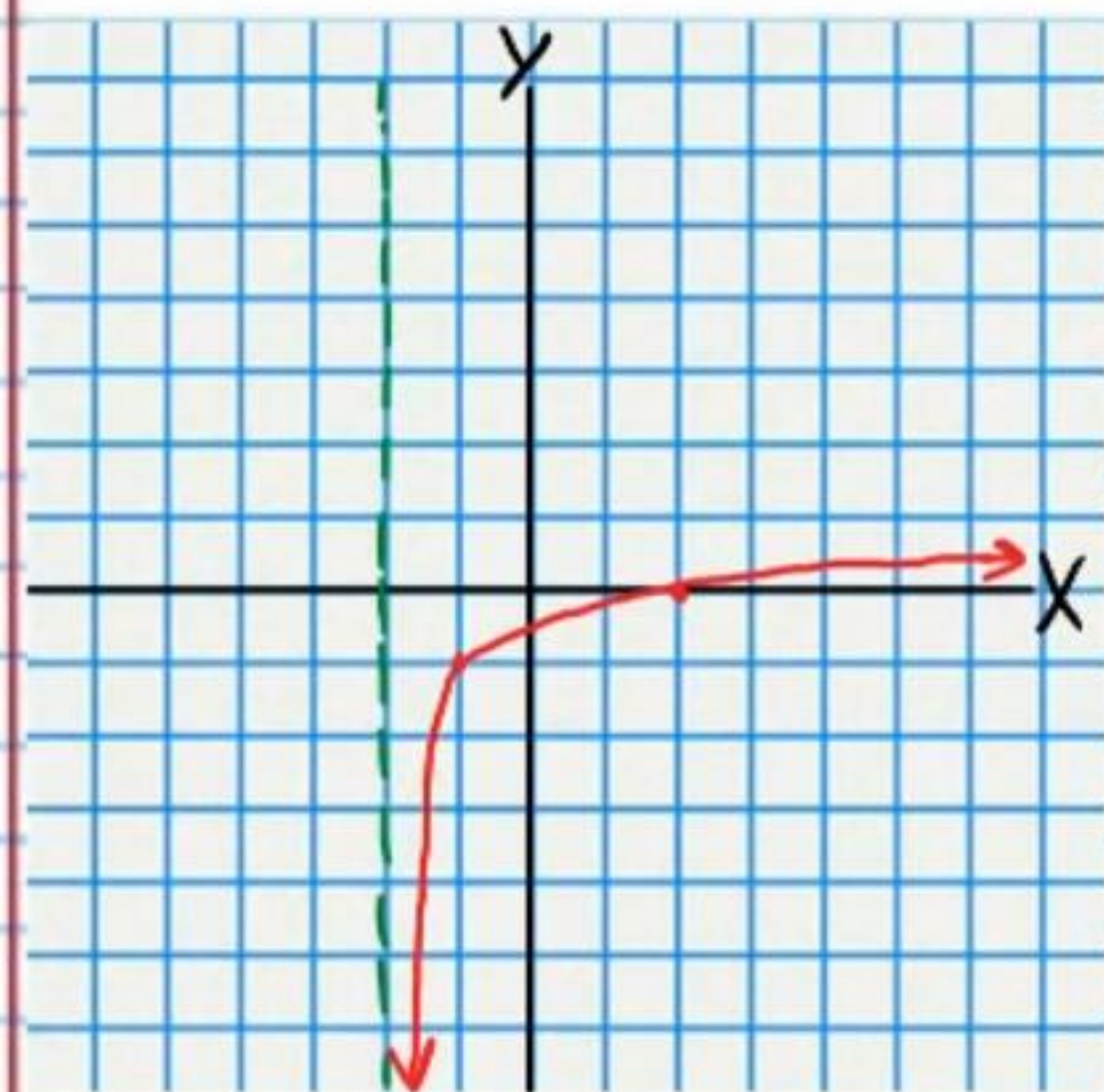


⑧ Logarithmic, $y = \log_3 x$

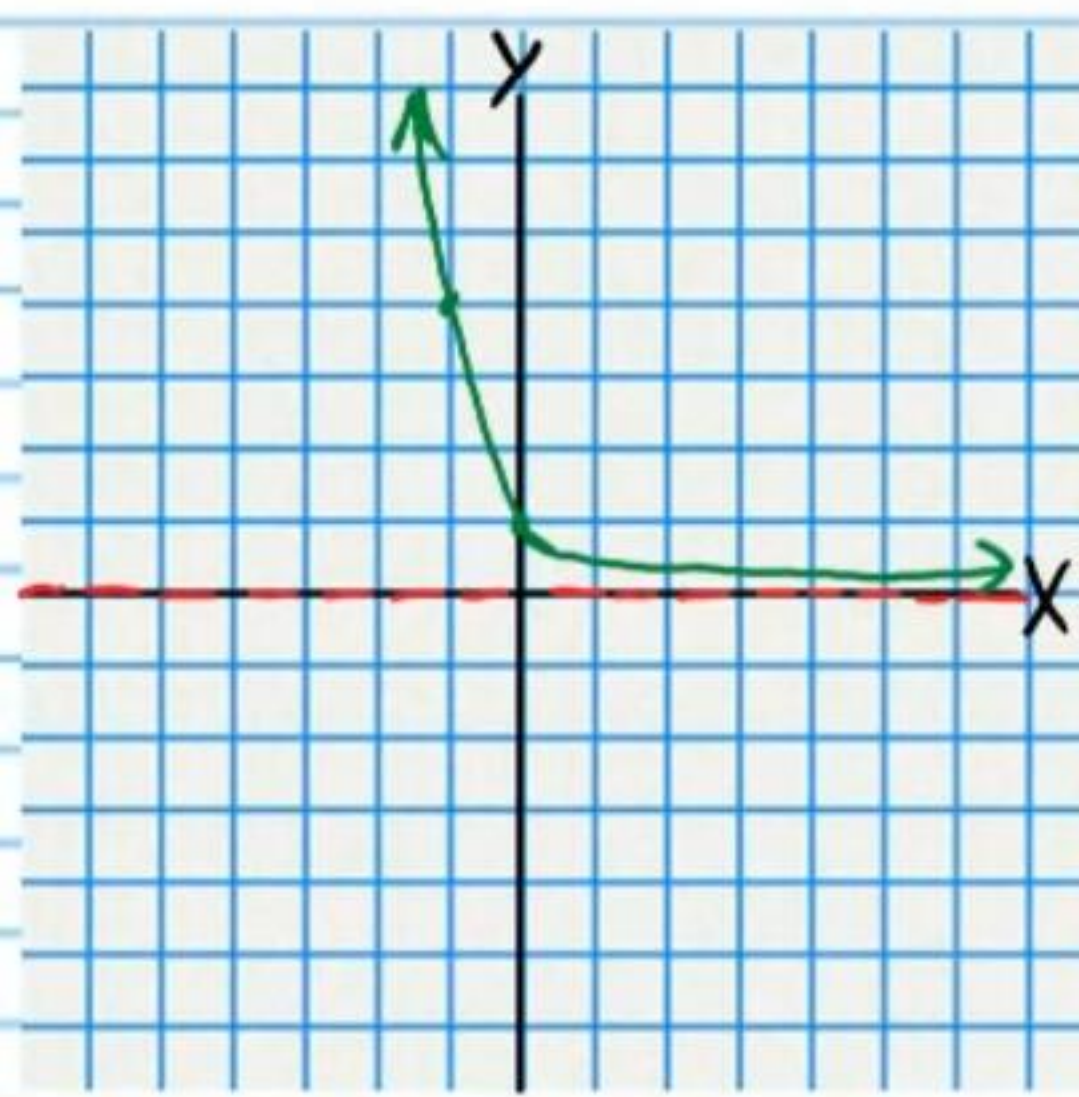
$x = 4$



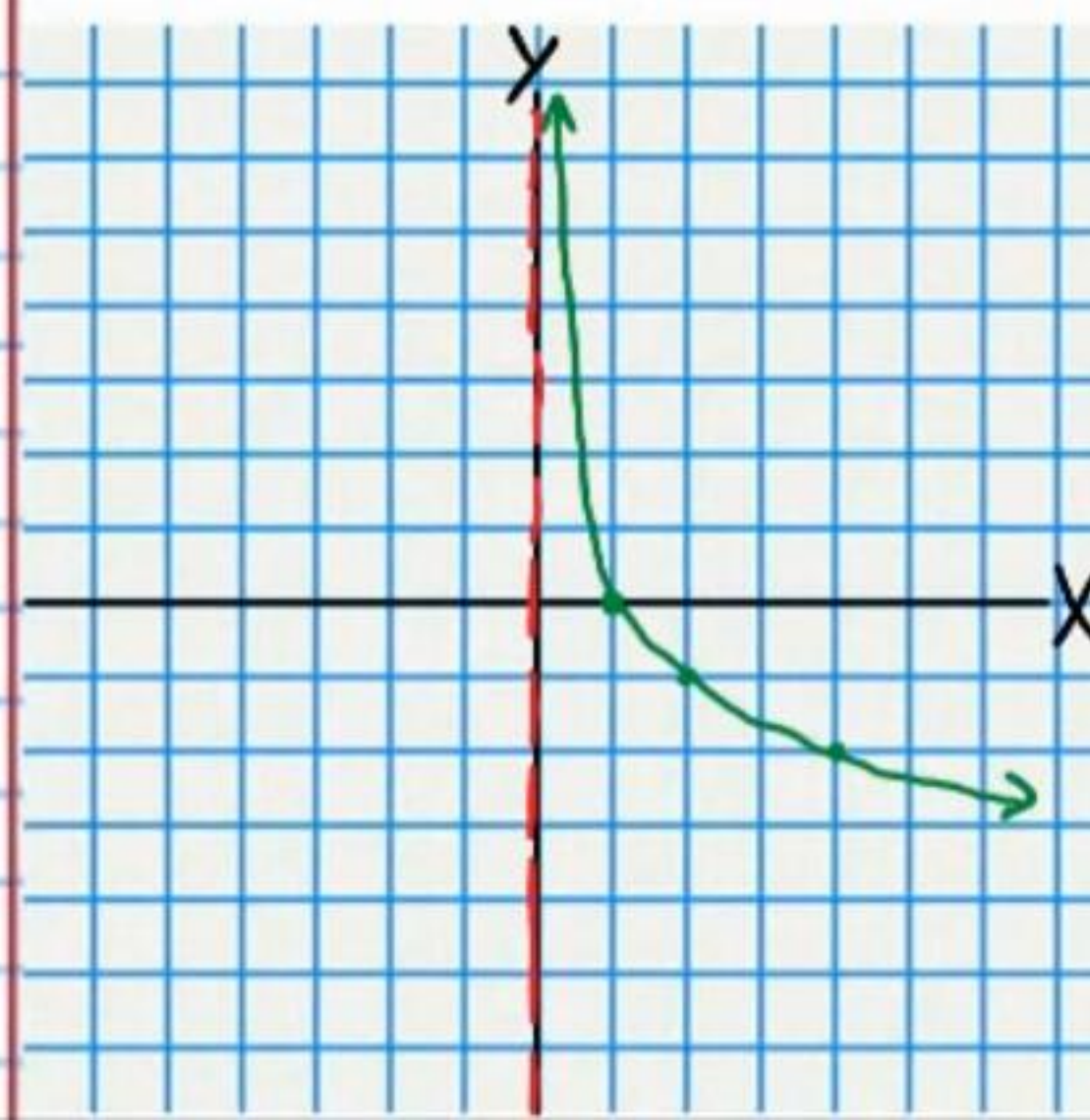
⑨ Logarithmic, $\log_4 X = Y$
 $X = -2$



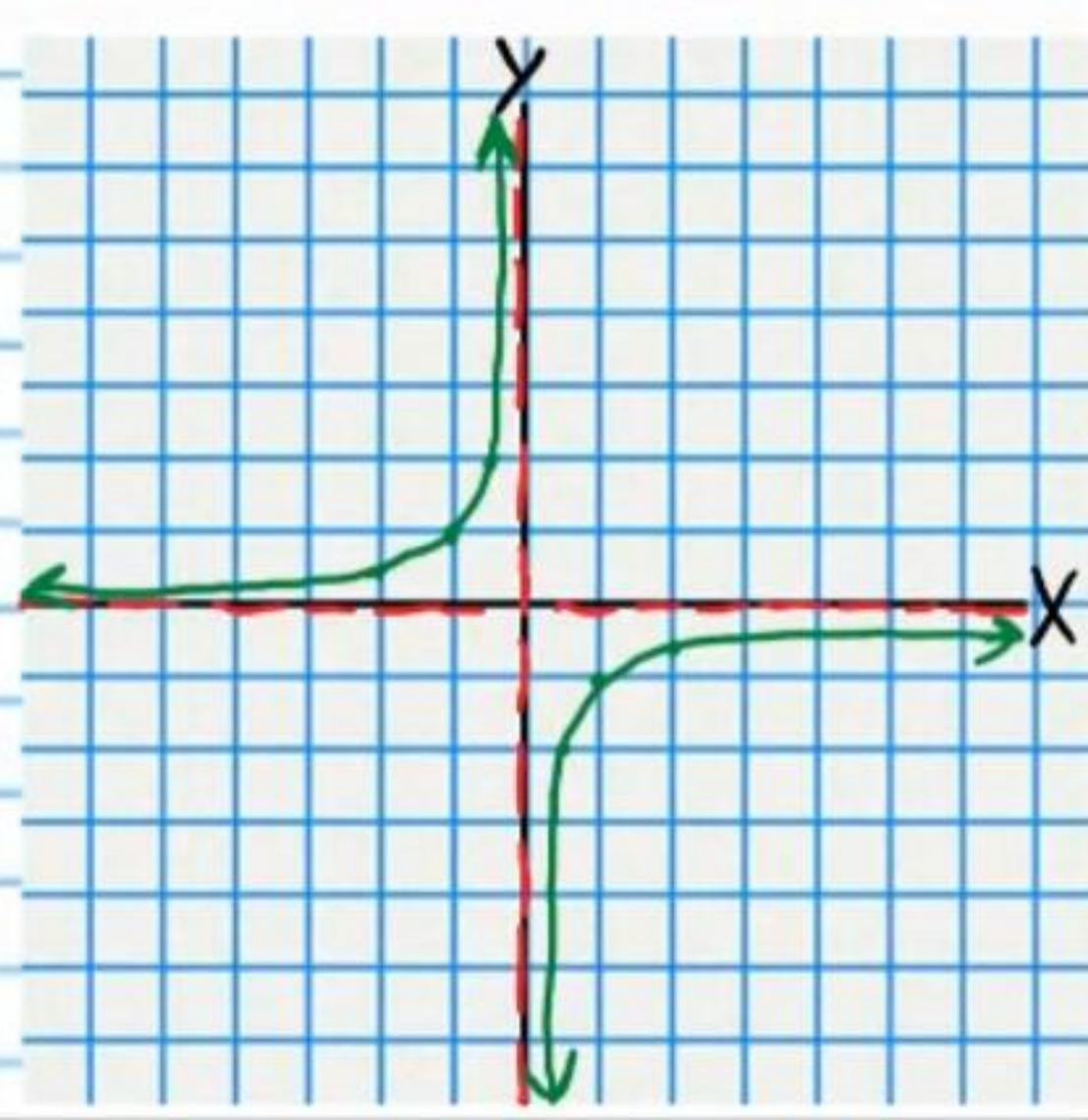
⑩ Exponential, $y = 4^x$
 $y = 0$



⑪ Logarithmic, $\log_2 X = Y$
 $X = 0$

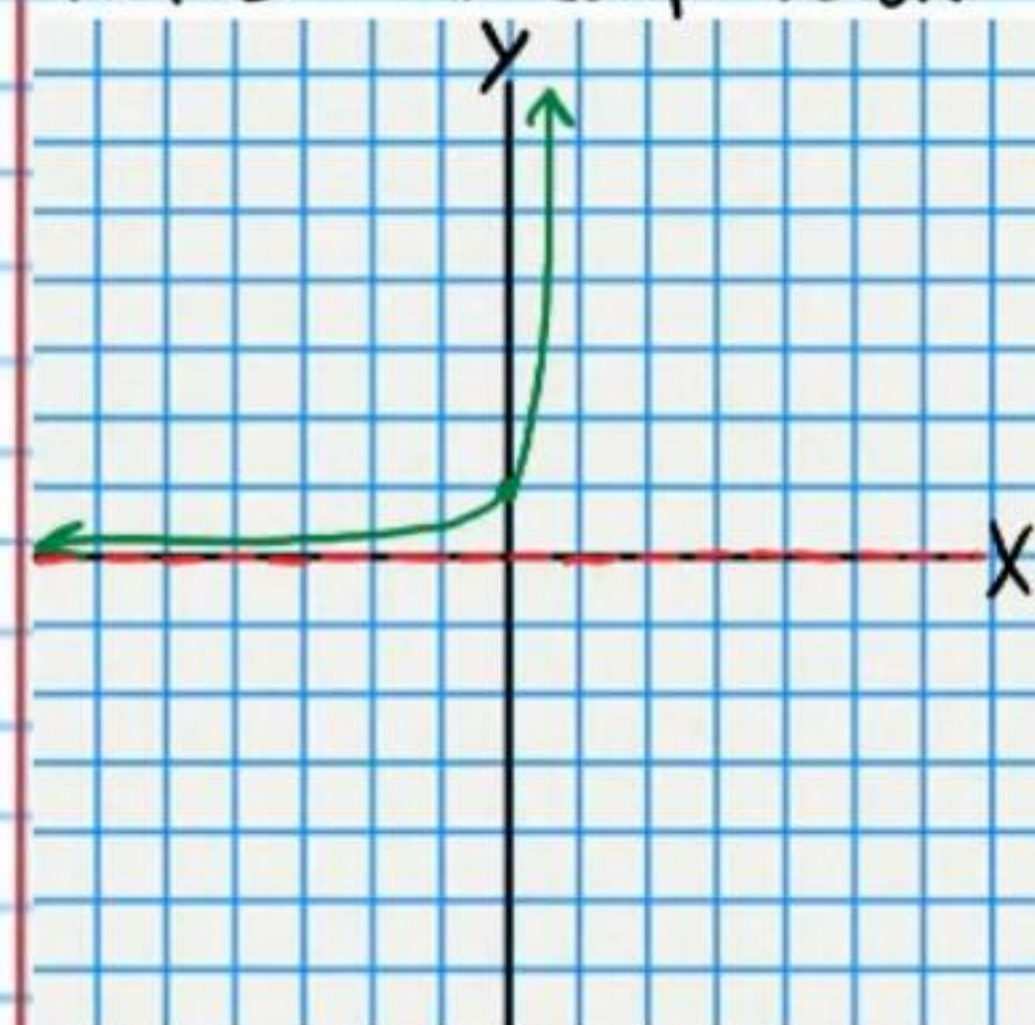


⑫ Reciprocal, $y = \frac{1}{X}$
 $X = 0, y = 0$



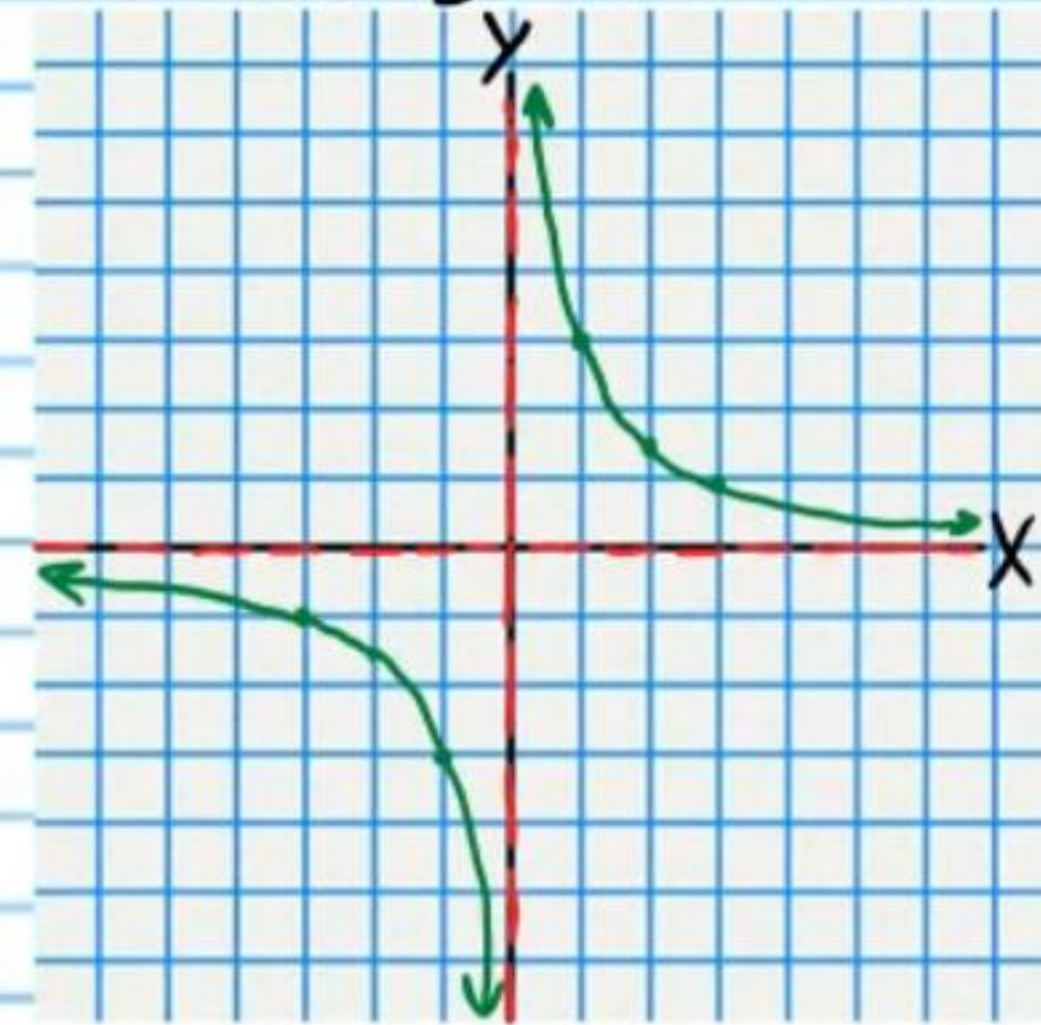
⑬ Exponential, $y = 2^x$
 $y = 0$

Horizontal Compression



⑭ Reciprocal, $y = \frac{1}{X}$
 $X = 0, y = 0$

Vertical Stretch



⑮ Logarithmic, $y = \log_2 X$
 $X = 0$

Vertical Compression

